



BERLIN CAUSAL INFERENCE WORKSHOP

Foundations of Difference-in-Differences

Day 1: The 2×2 , Parallel Trends, and Event Studies

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Sant'Anna Workshop on Causal Inference · Pisa

Who am I?

Scott Cunningham · Professor of Economics, Baylor University

- Empirical work: sex work, relationship markets, drug policy, abortion, corrections, suicide and mental illness — and AI agents and AI methodologies
- Podcast: *The Mixtape with Scott*
- Substack: *Scott's Substack*
- Teaching platform: *Mixtape Sessions*

Day 1

9:00am – 2:30pm DiD intro and covariates

- Canonical 2×2 difference-in-differences
- Conditional trends as functions of X_i
- Estimation under conditional parallel trends: regression adjustment, IPW, doubly-robust
- Sub-group heterogeneous treatment effects
- Coding lab: covariates

2:30pm – 5:00pm Continuous DiD and coding lab

- DiD where the treatment is a non-negative “dose” (continuous DiD)
- Treatment effects under a dose: total effect and per-unit effect
- Interpretation difficulties: the dose-specific ATT
- Estimation of continuous DiD in 2×2 settings
- Coding lab: continuous DiD

Day 2

9:00am – 2:00pm Staggered treatment timing

- Staggered timing and the definition of effects: $ATT(g, t)$
- Aggregation of group-time treatment effects
- Callaway and Sant'Anna estimator for $ATT(g, t)$
- Imputation-based approaches

2:00pm – 5:00pm Claude Code and DiD

- Claude Code introduction and Skills (but it seems down for me)
- Claude Code DiD Skills and example

Today: the basics of DiD

Four averages and three subtractions.

Calculation.

Identification.

Regressions.

Falsifications.

Today: the basics of DiD

Covariates

TWFE

Outcome regression

Propensity scores

Doubly robust

Continuous treatments

TWFE

Target parameters

Identification

Estimation

Staggered adoption

Various estimators

Target parameters

Identification

Estimation

Synthetic control methods

Abadie's original

Bias adjustment for poor fit

Synthetic diff-in-diff

Tomorrow

Claude Code

To be determined due to some outage (for me)

LESSON

1. DiD's Empirical Roots

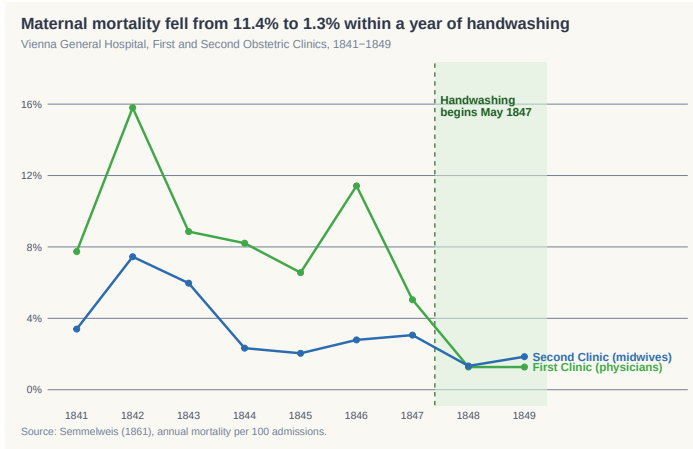
Vienna, 1840s: two clinics, two mortality rates

- Vienna General Hospital, First Obstetric Clinic
- Pregnant women were assigned to the *physicians' wing* or the *midwives' wing* by day of admission — effectively at random
- Physicians' wing: **13–18% mortality** from puerperal fever
- Midwives' wing: **about 3%**
- Women would deliver in the street rather than risk the physicians' wing

His hypothesis, and his DiD

- Semmelweis — an assistant on the wards — noticed the physicians did cadaver dissections before delivering babies. The midwives did not.
- Hypothesis: something on the physicians' hands was causing the fever.
- May 1847: he instituted **chlorinated lime handwashing for physicians only**
- Midwives served as the untreated comparison group — a DiD by hand

The data



Monthly mortality, First Obstetric Clinic. Handwashing began mid-1847.

Princeton, 1970s: Orley Ashenfelter

The Industrial Relations Section was founded 1922 — rigorous empirical labor research, the policy edge of US economics.

Orley Ashenfelter's leadership

- Long-time director. Set the tone: high-quality data, careful design, communicate to bureaucrats.
- Trained a cohort: **David Card**, **Josh Angrist**, **Bob LaLonde**, **Janet Currie**, Pischke, Dinardo, Case, others.
- Hired **Alan Krueger** as a junior colleague.

By the late 1970s, empirical labor was in trouble

- Different methods on the same question disagreed wildly
- The Princeton response: **design over modeling — estimators that mimic an RCT as closely as possible**
- Worker surveys, federal job training, massive panel regressions
- Audience was bureaucrats

Their eyes would glaze over whenever you said ‘regression.’

— Orley Ashenfelter

youtu.be/WnB3EJ8K7lg (3:53)

LESSON

2. DiD is Two Things: a Calculation and an Assumption

DiD is a research design, not a regression

A research design is an identification strategy: it reveals the invisible target parameter hidden inside a calculation.

1.

A calculation

(the 2×2)

+

2.

An assumption

(parallel trends)

The individual treatment effect

$$\delta_i = Y_i(1) - Y_i(0)$$

- One per unit. **Different** across units in general — treatment effects are heterogeneous.
- We only ever see one of $Y_i(1)$ or $Y_i(0)$. The other is the counterfactual.
- So δ_i is never directly observed at the individual level.

The ATT: a causal parameter has three ingredients

$$\text{ATT} = \underbrace{\mathbb{E}}_{\text{weighted expectations}} \left[\underbrace{Y(1) - Y(0)}_{\text{potential outcomes}} \mid \underbrace{D = 1, \text{Post}}_{\text{population}} \right]$$

- **Potential outcomes** — the contrast $Y(1) - Y(0)$
- **Population** — on whom: treated units, post-period
- **Weighted expectations** — how we average across that population ($\mathbb{E}[\cdot]$ defaults to equal weights)

The switching equation: you see one half

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

- For each unit: $Y_i(1)$ if treated, $Y_i(0)$ if not.
- The ATT asks about *both* columns. The data hands you one.

The rest of the day: strategies for recovering the ATT from the half you see.

LESSON

3. Five Regres- sions, One Number

Cheng & Hoekstra (2013): castle doctrine and homicide

The reform. “Castle doctrine” laws expanded the legal right to use lethal force in self-defense, removing the duty to retreat. Many US states passed expansions from **2005–2010**.

- **Panel:** all US states $\times$ years (state-year).
- **Outcome:** log homicide rate, FBI Supplementary Homicide Reports.
- **Treatment:** state passed an expansion of castle doctrine.
- **Comparison:** states that did not pass (never-treated).

For today: a clean 2×2 on **2005–2006**. Treated = states adopting in 2006; control = states that hadn't. *Cheng & Hoekstra (2013), “Does Strengthening Self-Defense Law Deter Crime or Escalate Violence? Evidence from Expansions to Castle Doctrine.” J. Human Resources 48(3): 821–854.*

The four cells, by hand on castle.dta

```
use castle.dta, clear
keep if year >= 2005 & year <= 2006
gen post = (year == 2006)
gen treat = (effyear == 2006)

summarize l_homicide if treat==1 & post==1 // Y11
summarize l_homicide if treat==1 & post==0 // Y10
summarize l_homicide if treat==0 & post==1 // Y01
summarize l_homicide if treat==0 & post==0 // Y00
```

$$\widehat{\text{DiD}} = (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00}) \approx \mathbf{0.10}$$

Log homicide rate ~ 10% higher in 2006-adopting states relative to non-adopters.

Regression 1: OLS with $\text{treat} \times \text{post}$

```
reg l_homicide post##treat, cluster(sid)
```

The coefficient on `1.post#1.treat` is the DiD.

$$Y = \beta_0 + \beta_1 \text{Post} + \beta_2 \text{Treat} + \beta_3 (\text{Post} \times \text{Treat}) + \varepsilon$$

The saturated specification. $\hat{\beta}_3 \approx 0.10$.

Regression 2: two-way fixed effects

```
xtset sid year  
xtreg l_homicide c.treat#c.post i.year, fe vce(cluster sid)
```

State fixed effects absorb time-invariant level differences. Year fixed effects absorb common shocks.

In the 2×2 case, the TWFE coefficient equals the DiD. Same number.

Regression 3: first-difference ΔY_i on D_i

```
keep sid year l_homicide treat  
reshape wide l_homicide, i(sid) j(year)  
gen dY = l_homicide2006 - l_homicide2005  
reg dY treat, vce(cluster sid)
```

$$\Delta Y_i = \alpha + \beta_{DD} D_i + u_i$$

- Collapse the panel: one ΔY per unit.
- Time-differencing wipes out unit fixed effects mechanically.
- Slope on treat = the DiD. Same number.

Regression 4: $(Y_{\text{treat}} - Y_{\text{ctrl}})$ on the post dummy

At each time t , take the across-group difference $D_t = \bar{Y}_{1,t} - \bar{Y}_{0,t}$. Then regress on a post indicator:

$$D_t = \alpha + \beta_{DD} \text{Post}_t + v_t$$

```
collapse (mean) l_homicide, by(treat year)
reshape wide l_homicide, i(year) j(treat)
gen Dt = l_homicide1 - l_homicide0
gen post = (year == 2006)
reg Dt post
```

- Regression 3 differences over *time*; Regression 4 differences over *groups* first.
- By Frisch–Waugh–Lovell, partialling out the two-way means runs in either order. Same number.

Regression 5: long differences — the Snow move

Compute one number per group, post minus pre, then subtract:

$$\widehat{\text{DiD}} = (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00})$$

- No regression. Two long differences. One subtraction between them.
- This is what John Snow did in 1854 with the water-company data.
- Identical to Regressions 1–4. **The arithmetic was the point all along.**

Five regressions, one number

Specification	What it does	$\hat{\beta}_{DD}$
1. OLS with <code>post##treat</code>	saturated interaction	≈ 0.10
2. Two-way fixed effects	absorb unit + time means	≈ 0.10
3. ΔY_i on D_i (first difference)	difference over time, then on D	≈ 0.10
4. D_t on Post_t (across-group diff)	difference over groups, then on Post	≈ 0.10
5. Long differences (Snow)	two means, one subtraction	≈ 0.10

*Same data, five specifications, identical to machine epsilon.
The 2×2 is the object; the regressions are just five ways to ask for it.*

LESSON

4. Decomposing the 2×2

The proof: add a zero, rearrange (five steps)

1. Start with the 2×2 :

$$\widehat{\text{DiD}} = [\mathbb{E}[Y \mid 1, \text{Post}] - \mathbb{E}[Y \mid 1, \text{Pre}]] - [\mathbb{E}[Y \mid 0, \text{Post}] - \mathbb{E}[Y \mid 0, \text{Pre}]]$$

2. Switching equation. Treated-post is $Y(1)$; the other three cells are $Y(0)$:

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}]] - [\mathbb{E}[Y(0) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]]$$

3. Add a zero. Insert $+\mathbb{E}[Y(0) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Post}]$:

$$\begin{aligned} \widehat{\text{DiD}} &= \mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Post}] \\ &\quad + \mathbb{E}[Y(0) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}] - [\mathbb{E}[Y(0) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]] \end{aligned}$$

4. Rearrange. First two terms = ATT; remainder = difference in $Y(0)$ trends:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1) - Y(0) \mid 1, \text{Post}]}_{\text{ATT}} + \underbrace{\{\Delta \mathbb{E}[Y(0) \mid 1] - \Delta \mathbb{E}[Y(0) \mid 0]\}}_{\text{non-parallel-trends bias}}$$

5. Read. If the bias equals zero (*parallel trends*), then $\widehat{\text{DiD}} = \text{ATT}$.

The bias term *is* selection bias — on trends, not levels

The “non-parallel-trends bias” isn’t a new concept. It’s selection bias, in a different costume.

$$\text{bias} = \underbrace{\Delta \mathbb{E}[Y(0) \mid D=1]}_{\text{how treated's } Y(0) \text{ moves}} - \underbrace{\Delta \mathbb{E}[Y(0) \mid D=0]}_{\text{how controls' } Y(0) \text{ moves}}$$

- **Cross-section selection bias:** are treated and control different in *levels* of $Y(0)$?
- **DiD’s selection bias:** are treated and control different in *trends* of $Y(0)$?
- Different levels are fine. Unit fixed effects soak them up.
- What can’t differ is the *rate* at which $Y(0)$ moves across groups.

The trade DiD makes vs. randomization

RCTs: treated and controls identical in everything — levels and trends.

DiD: identical only in *trends* of $Y(0)$. **Cheaper assumption. Weaker identification.**

Case 1: NA holds, never-treated comparison

Switching equation. Treated-post is $Y(1)$; the other three cells are $Y(0)$.

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Pre}]] - [\mathbb{E}[Y(0)|0, \text{Post}] - \mathbb{E}[Y(0)|0, \text{Pre}]]$$

Add the zero. Insert $+\mathbb{E}[Y(0)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Post}]$. Rearrange:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1) - Y(0)|1, \text{Post}]}_{\text{ATT}} + \underbrace{\{\Delta\mathbb{E}[Y(0)|1] - \Delta\mathbb{E}[Y(0)|0]\}}_{\text{non-PT bias} = 0 \text{ under PT}}$$

Numerical example

$$\text{ATT} = 10, \text{PT bias} = 0. \quad \widehat{\text{DiD}} = 10. \quad \checkmark$$

Case 2: NA fails — anticipation contaminates Y_{Pre}

Suppose the treated group anticipates and adjusts *before* the policy lands. Their pre-period outcome contains some $Y(1)$ response: $\mathbb{E}[Y|1, \text{Pre}] = \mathbb{E}[Y(0)|1, \text{Pre}] + \text{ATT}_{\text{pre}}$.

Same +0 trick, propagated:

$$\widehat{\text{DiD}} = \underbrace{\text{ATT}_{\text{Post}}}_{\text{what we want}} + \underbrace{\text{non-PT bias}}_{=0} - \underbrace{\text{ATT}_{\text{pre}}}_{\text{anticipation}}$$

Numerical example

$$\text{ATT} = 10, \text{PT bias} = 0, \text{ATT}_{\text{pre}} = 3. \quad \widehat{\text{DiD}} = 10 - 3 = 7. \text{ biased down}$$

If treatment effects are roughly constant, DiD can collapse to zero.

Case 3: the control group also receives treatment in Post

Now the comparison group's Post outcome is also $Y(1)$, not $Y(0)$. Switching equation:

$$\widehat{\text{DiD}} = \left[\mathbb{E}[Y(1)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Pre}] \right] - \left[\mathbb{E}[Y(1)|0, \text{Post}] - \mathbb{E}[Y(0)|0, \text{Pre}] \right]$$

Add two zeros — one missing counterfactual on each side — and rearrange:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1) - Y(0)|1, \text{Post}]}_{\text{ATT}} + \underbrace{\left\{ \Delta \mathbb{E}[Y(0)|1] - \Delta \mathbb{E}[Y(0)|0] \right\}}_{\text{non-PT bias} = 0 \text{ under PT}} - \underbrace{\mathbb{E}[Y(1) - Y(0)|0, \text{Post}]}_{\text{ATT}_{D=0, \text{Post}} \text{ (spillover)}}$$

Numerical example

$\text{ATT} = 10$, $\text{PT bias} = 0$, $\text{ATT}_{D=0, \text{Post}} = 4$. $\widehat{\text{DiD}} = 10 - 4 = 6$. **biased toward zero**

Case 4: the comparison group is *always* treated

The comparison group was treated *before* our data starts. Both their Pre and Post outcomes are $Y(1)$, not $Y(0)$. Switching equation:

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Pre}]] - [\mathbb{E}[Y(1)|0, \text{Post}] - \mathbb{E}[Y(1)|0, \text{Pre}]]$$

Add two zeros — $\pm \mathbb{E}[Y(0)|0, \text{Post}]$ and $\pm \mathbb{E}[Y(0)|0, \text{Pre}]$ on the comparison side — and rearrange:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1) - Y(0)|1, \text{Post}]}_{\text{ATT}} + \underbrace{\{\Delta \mathbb{E}[Y(0)|1] - \Delta \mathbb{E}[Y(0)|0]\}}_{\text{non-PT bias} = 0 \text{ under PT}} - \underbrace{\Delta \text{ATT}_{D=0}}_{\text{change in comparison TE}}$$

Numerical example

$\text{ATT} = 10$, $\text{PT bias} = 0$, $\Delta \text{ATT}_{D=0} = 2$. $\widehat{\text{DiD}} = 10 - 2 = 8$. **biased toward zero**

Four cases, four failures of the punchline. The +0 trick still works — only the zero you add changes.

Summary: four ways the 2×2 decomposes

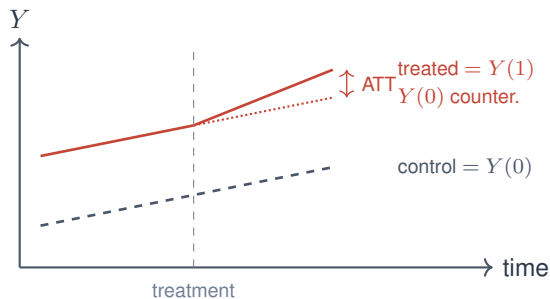
What holds	$\widehat{\text{DiD}} =$
NA, never-treated control	$\text{ATT} + \text{PT-bias}$
PT, no NA, never-treated control	$\text{ATT}_{\text{Post}} - \text{ATT}_{\text{Pre}}$
PT, NA, always-treated control	$\text{ATT} - \Delta \text{ATT}_{D=0}$
PT, NA, SUTVA fails (spillover)	$\text{ATT} - \text{ATT}_{D=1, \text{Post}}$

All four derivable the same way: write down the 2×2 , apply the switching equation, add the right series of zeros.

LESSON

5. Card and Krueger: Parallel Trends, Visualized

Parallel trends, drawn



$$\mathbb{E}[Y(0) \mid D=1, \text{Post}] - \mathbb{E}[Y(0) \mid D=1, \text{Pre}] = \mathbb{E}[Y(0) \mid D=0, \text{Post}] - \mathbb{E}[Y(0) \mid D=0, \text{Pre}]$$

*A 2×2 of $Y(0)$ terms. **If it isn't four $Y(0)$ s, it isn't the PTA.***

Under PT, the 2×2 hits the ATT

With (i) **No Anticipation**, (ii) **Parallel Trends in $Y(0)$** , and (iii) a clean (never-treated) comparison group:

$$\text{ATT}_{\text{post}} = \underbrace{\mathbb{E}[\Delta Y \mid D = 1]}_{\text{treated change}} - \underbrace{\mathbb{E}[\Delta Y \mid D = 0]}_{\text{control change}}$$

- The 2×2 calculation equals the population's ATT.
- Four averages. Three subtractions. One target parameter.

This is the punchline. Everything that follows is variations on this theme.

New Jersey raises its minimum wage. Pennsylvania doesn't.

- **April 1992**: New Jersey raises its minimum wage from \$4.25 to \$5.05 — a 19% increase.
- Pennsylvania, just over the river, stays at \$4.25.
- Card and Krueger survey 410 fast-food restaurants on both sides of the border, before (Feb 1992) and after (Nov 1992).
- The natural 2×2 : **treated = NJ, control = PA, pre/post the policy.**

Card & Krueger, American Economic Review, 1994.

What Card and Krueger *found*

Employment per restaurant	Pre (Feb 1992)	Post (Nov 1992)	Change
NJ (treated)	20.44	21.03	+0.59
PA (control)	23.33	21.17	−2.16
DiD (NJ – PA)			+2.76 jobs/restaurant

- Employment in NJ fast food *rose* slightly, relative to PA, after the wage hike.
- Standard error puts the effect roughly indistinguishable from zero — but the sign is the opposite of what theory predicted.
- **No measurable disemployment effect.** The textbook had failed a real test.

The reaction was personal

No self-respecting economist would claim that increases in the minimum wage increase employment. Such a claim becomes equivalent to a denial that there is even minimal scientific content in economics.

— James Buchanan (Nobel 1986), *WSJ*, April 1996

It cost me a lot of friends. People that I had known for many years became very angry or disappointed. They thought that in publishing our work we were being traitors to the cause of economics as a whole.

— David Card, interview

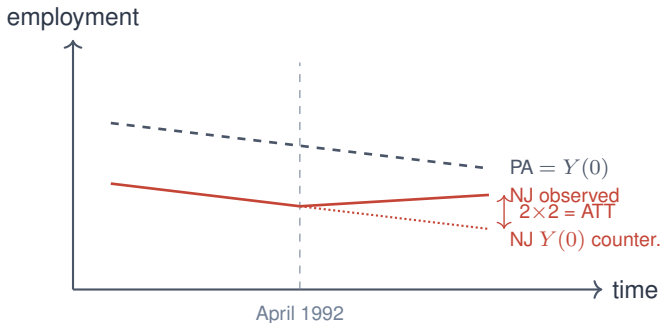
The price of a credible empirical paper that contradicted the consensus. Krueger paid more.

Hear it from them — three short interview clips

- **Orley Ashenfelter on the profession's reaction** (start 31:20)
youtu.be/M0tbuRX4eyQ?t=1882
- **Card on the hostility from peers** (1:07–1:10)
youtu.be/1soLdywFb_Q?t=4020
- **Card on what happened to Alan Krueger** (1:16–1:17)
youtu.be/1soLdywFb_Q?t=4556

The DiD literature did not arrive politely. It was earned through paper-by-paper combat, much of it personal.

Parallel trends visualized: the control *is* the counterfactual



The dotted line is what NJ would have done absent the policy. We never see it.

PT lets PA's slope stand in for it.

Aside: there's also an *across-group* PT we never name

The usual statement (within-group, across-time)

The comparison group's $Y(0)$ trend stands in for what the treated group's $Y(0)$ trend would have been.

The same assumption, stated from the other side (across-group, within-time)

The treated group's $Y(0)$ trend (had they not been treated) would equal the comparison group's $Y(0)$ trend.

- Two phrasings of one assumption: $Y(0)$ trends are common across groups.
- Event-study β_k vs $t = -1$ computes the within-group across-time piece.
- Regression 3 (ΔY on D) and Regression 4 (D_t on Post) are FWL mirrors — same PT from the two directions.

ASIDE — drop in or drop out if behind schedule.

Pre-trends are the gut check — not a proof of PT

- Card–Krueger held up because in the pre-period, NJ and PA fast-food employment moved together.
- If NJ employment had already been diverging before April 1992, the +2.76 estimate would have been mostly trend, not policy.
- Flat pre-trends → evidence (not proof) the design holds.
- **The 2×2 was credible because of the pre-period parallelism — not because of the regression coefficient.**

More later (Roth, Roth–Sant’Anna): when pre-trends look flat but the test still over-rejects.

LESSON

6. Open the Sheet

Open the Sheet: stability and trends, by hand

Live demo

Compute the four cells, three subtractions, and the by-hand event study — in a spreadsheet.

docs.google.com/spreadsheets/d/1onabpc14JdrGo6NFv0zCWo-nuWDLLV2L1qNogDT9SBw

Four tabs in sequence: Potential Outcomes $\rightarrow$ 2X2 $\rightarrow$ Balancing $\rightarrow$ 2XT #1.

LESSON

7. Standard Errors

Three flavors of standard error

- **Vanilla (homoskedastic)**: assumes $\text{Var}(\varepsilon_i \mid X_i) = \sigma^2$ for all i .
- **Robust (Eicker–Huber–White, HC)**: each residual speaks for itself — handles heteroskedasticity.
- **Cluster-robust (CRVE)**: groups residuals by cluster; within-cluster errors correlate freely.

Same point estimate, three different SEs. The SE you pick is a claim about the error structure.

HC1 numerically: each $\hat{e}_i^2$ weighted by leverage

Six observations. Three pairs at $\tilde{x} \in \{-2, 0, +2\}$. Fitted $\hat{\beta}_1 = 1$; residuals shown.

i	cluster g	$\tilde{x}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$\tilde{x}_i^2 \cdot \hat{e}_i^2$
1	1	-2	+0.5	0.25	1.00
2	1	-2	+0.5	0.25	1.00
3	2	0	-1.0	1.00	0.00
4	2	0	-1.0	1.00	0.00
5	3	+2	+0.5	0.25	1.00
6	3	+2	+0.5	0.25	1.00
meat =					4.00

With $n = 6$, $k = 2$, and $S_{xx} = \sum_i \tilde{x}_i^2 = (-2)^2 + (-2)^2 + 0 + 0 + 2^2 + 2^2 = 16$:

$$\widehat{\text{Var}}_{\text{HC1}}(\hat{\beta}_1) = \frac{n}{n-k} \cdot \underbrace{\frac{1}{S_{xx}}}_{\text{bread}} \cdot \underbrace{\text{meat}}_{4.00} \cdot \underbrace{\frac{1}{S_{xx}}}_{\text{bread}} = \frac{6}{4} \cdot \frac{4.00}{16 \cdot 16} = \frac{6}{4} \cdot \frac{4.00}{256} = 0.02344$$

$$\Rightarrow \text{SE}_{\text{HC1}} = \sqrt{0.02344} \approx 0.153$$

Obs 3–4 contribute zero because $\tilde{x} = 0$ kills the leverage weight. **HC1 treats all 6 obs as independent.**

Why HC-robust barely helps in DiD

The HC correction reweights residuals by $X_i X_i'$. In a DiD with binary $D_i \in \{0, 1\}$ and roughly balanced groups:

- Treated and control units have similar *leverage*: each side carries $\approx$ half the weight in $X'X$.
- Residual variance is roughly symmetric across the two groups (binary D doesn't generate a fan shape in $\hat{e}^2$).
- The weighted sum $\sum_i \hat{e}_i^2 X_i X_i'$ ends up not very different from $\hat{\sigma}^2(X'X)$.

Consequence

$SE_{HC} \approx SE_{vanilla}$ for typical DiD specifications.

The over-rejection problem is not heteroskedasticity. It's something else.

CRVE numerically: same 6 obs, group scores, then square

Same 6 observations. Now compute $s_g = \sum_{i \in g} \tilde{x}_i \hat{e}_i$ for each cluster, then s_g^2 :

i	g	$\tilde{x}_i$	$\hat{e}_i$	$\tilde{x}_i \hat{e}_i$			
					g	s_g	s_g^2
1	1	-2	+0.5	-1			
2	1	-2	+0.5	-1	1	-2	4
3	2	0	-1.0	0	2	0	0
4	2	0	-1.0	0	3	+2	4
5	3	+2	+0.5	+1	meat =		8
6	3	+2	+0.5	+1			

With $G = 3$, $n = 6$, $k = 2$, and $S_{xx} = \sum_i \tilde{x}_i^2 = 16$, so $S_{xx}^2 = 256$:

$$\widehat{\text{Var}}_{\text{CRVE}}(\hat{\beta}_1) = \frac{G}{G-1} \cdot \frac{n-1}{n-k} \cdot \frac{\text{meat}}{S_{xx}^2} = \frac{3}{2} \cdot \frac{5}{4} \cdot \frac{8}{256} = 0.05859 \Rightarrow \text{SE}_{\text{CRVE}} \approx 0.242$$

HC1: 0.153 vs. **CRVE:** 0.242 (**58% larger**). Same data. The cluster cross-terms changed the *meat* from 4 to 8.

Bertrand, Duflo & Mullainathan (QJE 2004): the placebo-law test

“How Much Should We Trust Differences-in-Differences Estimates?”

- BDM generate *placebo laws*: assign random treatment dates to states in a real CPS panel.
- True effect by construction: $\beta = 0$. So $\Pr(\text{reject } H_0)$ should equal $\alpha = 5\%$.
- **Vanilla SE**: rejection rate $\approx 45\%$. Wildly over-rejects.
- **Heteroskedasticity-robust (HC1)**: rejection rate *also* $\approx 45\%$. **HC barely moves the needle.**
- **Cluster-robust at state level**: rejection rate $\approx 5\%$. The fix.

Why HC doesn't save you

Binary, balanced D + within-state serial correlation. The het-correction has nothing to do with the actual problem. **Clustering is doing all the work.**

LESSON

8. Population Weighting

Ten people, two counties

Person	$Y(0)$	$Y(1)$	δ_i	County
Alan	0	1	1	1
Betty	1	1	0	1
Chad	1	1	0	1
Daniel	0	0	0	1
Edith	0	1	1	1
Frank	0	1	1	1
George	0	0	0	1
Hank	0	1	1	1
Ida	1	0	-1	2
Janet	1	0	-1	2

County 1: eight people. County 2: two people. Compute the average δ for each county, and across all ten. Same answer?

Same data, two weights, two parameters

- **Within counties:** County 1 avg $\delta = \frac{1}{8}(1+0+0+0+1+1+0+1) = +0.5$.
County 2 avg $\delta = \frac{1}{2}(-1+(-1)) = -1.0$.
- **ATE for the average person:** $\frac{1}{10}(8 \cdot 0.5 + 2 \cdot (-1)) = +0.2$.
- **ATE for the average county:** $\frac{1}{2}(0.5 + (-1)) = -0.25$.

*Same potential outcomes. **Different signs.***

Whichever you report depends on what you're trying to answer.

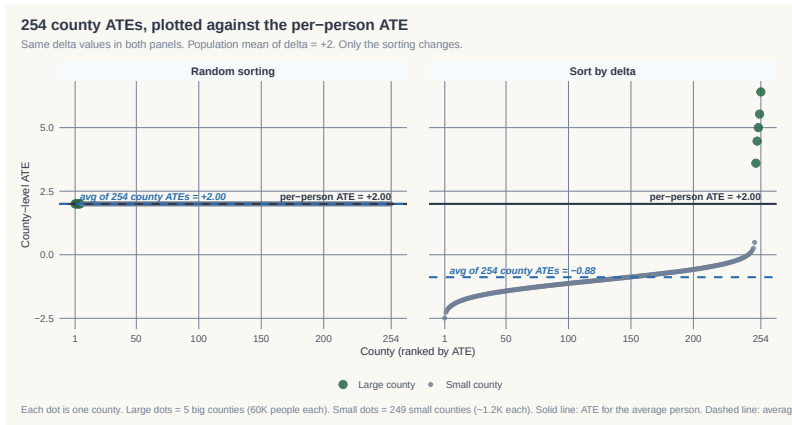
Two parameters, two answers

Scale the example up: **Texas, 254 counties, 30M people**. 5 biggest counties hold roughly half the population. Heterogeneous δ_i across people, with the high- δ folks concentrated in the 5 big counties and the low- δ folks in the 249 small ones.

- **Per-person ATE:** $\frac{1}{N} \sum_i \delta_i$ — weight every person by $1/N$. **Answer: +2.**
- **Avg of county ATEs:** $\frac{1}{254} \sum_c \bar{\delta}_c$ — weight every county by $1/254$. **Answer: -0.88.**

Same data. Two weights. Two parameters. Sometimes two signs.

Sorting on δ can flip the sign of your answer



Random: every county's ATE equals the population's. δ -sort: per-person line at +2, avg of county ATEs flips to -0.88 .

Random sorting is the benchmark — rarely reality. People choose where to live, and on traits that move δ .

The weighted mean is a covariance correction

For any outcome Y and any weight S :

$$\tilde{\mathbb{E}}[Y] \equiv \frac{\mathbb{E}[SY]}{\mathbb{E}[S]} = \mathbb{E}[Y] + \frac{\text{Cov}(S, Y)}{\mathbb{E}[S]}$$

- Weighted mean = unweighted mean + a **covariance correction**.
- $\text{Cov}(S, Y) > 0 \Rightarrow$ weighted exceeds unweighted; $< 0 \Rightarrow$ smaller.
- Ten-people table: weight by county size $S \in \{8, 2\}$ and compute $\tilde{\mathbb{E}}[\delta]$.

Stata

```
gen size = 8 if county==1
replace size = 2 if county==2
summarize delta [aweight=size]
* weighted mean = +0.2
summarize delta
* unweighted mean = -0.25
```

R

```
df$size <- ifelse(df$county==1, 8, 2)
weighted.mean(df$delta, df$size)
# weighted mean = +0.2
mean(df$delta)
# unweighted mean = -0.25
```

Pick a target parameter

- Three weights. Three signs. Three answers.
- The choice is about *which causal effect you actually want*.
- **Person-weighted** for aggregate human welfare: how many people are affected, by how much?
- **Unit-weighted** when locations have autonomy: each policy unit chooses independently — the unit *is* the agent.

Today's target: the ATT.

The average effect for the units that actually received the treatment.

LESSON

9. Event Studies: PT Made Falsifiable

Event studies are PT falsifications. What is a falsification?

- Popperian reasoning: you don't confirm a theory — you try to falsify it.
- A rival theory earns scrutiny by making a *prediction you can check*.
- Claim a confounder. Derive what *else* it would do. Test that.
- If the prediction holds → you haven't confirmed the rival, but you haven't ruled it out either.
- If the prediction fails → the rival weakens.

Popperian asymmetry. Failing to reject the rival does not confirm you. But rejecting the rival's prediction is real evidence.

Two forms of falsification

- **Same outcome, different group.** A group your treatment shouldn't reach, but the rival would.
Min wage on low-wage employment? Re-run on higher-wage workers.
- **Same group, different outcome.** An outcome your treatment shouldn't move, but the rival would.
Cigarette tax on smoking-related hospitalizations? Re-run on appendicitis and fractures.

PT falsifications come in both forms. Event studies are one (same group, different “outcome” — a period when no treatment happened).

Famous falsification 1: obesity contagion

- Christakis & Fowler (2007): **obesity is contagious** in social networks.
- Cohen-Cole & Fletcher (2008): re-run the same model on **height, acne, headaches** — same Framingham network, same identification.
- All three also show “contagion.”

We have not been able to become taller by having tall friends.

— Cohen-Cole & Fletcher (2008)

Same group, different outcome. The result isn't necessarily wrong — the empirical model is.

Famous falsification 2: milk addiction

- Becker–Murphy (1988): serial correlation in cigarette / alcohol $\Rightarrow$ “**rational addiction.**”
- Auld & Grootendorst (2004, *J. Health Econ.*): apply the same estimator to **milk, eggs, oranges.**
- All three look “rationally addictive.”

*Same group, different outcome. Nobody is rationally addicted to milk. **The identification — not the goods — is what's broken.***

Point: PT makes predictions beyond the event study. Placebo groups and placebo outcomes are the other two forms.

Parallel trends is about $Y(0)$, not “trends” abstractly

The parallel-trends assumption is a statement about an unobserved object:

$$\mathbb{E}[Y(0)_t - Y(0)_{t-1} \mid D=1] = \mathbb{E}[Y(0)_t - Y(0)_{t-1} \mid D=0]$$

- PT is the counterfactual untreated outcome's trend. **Not** a statement about OBSERVED Y trends in general.
- Observed Y for treated post-treatment is $Y(1)$, not $Y(0)$. The PT assumption can't be read off it.
- The confusion comes from saying “trends are parallel.” What we mean is “**untreated potential outcome trends are parallel.**”

Why this matters

We never see treated $Y(0)$ post-treatment. Pre-treatment data is the only window onto whether a similar assumption *could* hold post. That's where event studies come in.

Event studies: pre-treatment placebos for PT

The falsification logic. Suppose some confounder both shifts $Y(0)$ trends *and* is associated with treatment timing. That's a PT violation.

- If a similar confounder structure existed *before* treatment, we should see treated and control Y trends diverge in the pre-period too.
- Estimate placebo “effects” at $t = -2, -3, \dots, -q$ where there is no treatment. If those coefficients are large or trending, we've rejected the assumption that drives the post-period.

Popperian asymmetry

Flat pre-trends *fail to reject* PT — they do not prove it.

Sloped pre-trends *reject* the design.

Pre-treatment placebos are a falsification, not a proof.

An event study is the 2×2 viewed in motion

- Instead of one pre-period and one post-period, use *many*.
- Estimate a separate effect at each event-time relative to treatment.
- Plot them: $\hat{\beta}_{-q}, \dots, \hat{\beta}_{-2}, \hat{\beta}_{-1}, \hat{\beta}_0, \hat{\beta}_{+1}, \dots, \hat{\beta}_{+q}$.
- The graph tells two stories: **pre-trends** (left of zero) and **dynamics** (right of zero).

Same arithmetic as the 2×2 , repeated against a common reference period.

Every coefficient is a 2×2 against $t = -1$

For each event-time k (with $t = -1$ as the reference):

$$\hat{\beta}_k = (\bar{Y}_k^T - \bar{Y}_{-1}^T) - (\bar{Y}_k^C - \bar{Y}_{-1}^C)$$

- Two long differences, one subtraction — the 2×2 again, with $t = -1$ as the pre-period.
- Pre-period $\hat{\beta}_k$ ($k < -1$): asks whether trends were already diverging.
- Post-period $\hat{\beta}_k$ ($k \geq 0$): the dynamic ATTs.

Drop $k = -1$ as the reference. Multicollinearity won't let you keep all the dummies.

The saturated OLS specification

$$Y_{it} = \alpha_i + \tau_t + \sum_{t \neq t_0-1} \beta_t \cdot (D_i \cdot \mathbf{1}\{\text{year} = t\}) + \varepsilon_{it}$$

- α_i — unit fixed effects (absorb levels).
- τ_t — calendar-year fixed effects (absorb common shocks).
- One interaction per calendar year, omitting $t_0 - 1$ (the year just before treatment) as the reference.
- Each β_t is the 2×2 of the previous slide: (treated change from t_0-1 to t) — (control change from t_0-1 to t).

In a $2 \times T$ design the calendar year is the event-time. We save k for staggered treatment, where cohorts have different treatment dates.

Computing $\hat{\beta}_{2004}$ on castle.dta — by hand vs. OLS

Castle doctrine cohort treated in 2006. Use 2005 as the omitted reference. The 2004 coefficient is one event-study 2×2 .

By hand

```
su l_homicide if treat==1 & year==2004
* mean = 1.802504
su l_homicide if treat==1 & year==2005
* mean = 1.775398
su l_homicide if treat==0 & year==2004
* mean = 1.254353
su l_homicide if treat==0 & year==2005
* mean = 1.282884

di (1.802504-1.775398) - (1.254353-1.282884)
* 0.05564
```

$$\hat{\beta}_{2004} = \mathbf{0.05564}$$

Via OLS in Stata

```
reg l_homicide ///
    treat##ib2005.year, ///
    cluster(state)

di _b[1.treat#2004.year]
* 0.05564
```

$$\hat{\beta}_{2004} = \mathbf{0.05564}$$

Cluster-robust SE: 0.060. Match to 5 decimals.

Every event-study coefficient is a 2×2 relative to $t = -1$ (here, 2005).

LESSON

10. Five Pieces of a Strong DiD

Medicaid expansion and near-elderly mortality

Miller, Johnson & Wherry (2021), *Quarterly Journal of Economics*.

- Did the ACA Medicaid expansion (2014) reduce mortality among the **near-elderly** (ages 55–64)?
- Comparison: states that expanded vs. states that didn't, before vs. after 2014.
- Headline finding: $\approx 9.2\%$ **reduction** in near-elderly mortality.

Why *near-elderly*?

Everyone 65+ has Medicare — Medicaid expansion is irrelevant for them. The elderly become a natural **placebo group** living in the same economies.

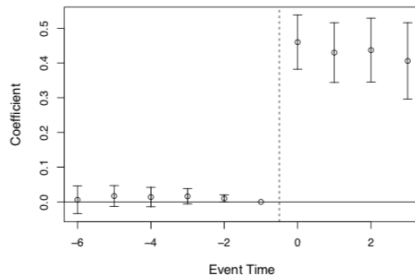
Today: walk through how they make the case using the five pieces of a strong DiD.

Five pieces of a strong DiD — start with bite

1. **Bite** — the policy actually changed what it was supposed to change.
2. **Event study** — pre/post coefficients around treatment.
3. **Falsification** — run a rival theory on a placebo group or outcome.
4. **Main result** — the DiD on the actual outcome.
5. **Mechanism** — the causal pathway from D to Y .

If the policy didn't actually move what it was supposed to move, nothing downstream is attributable to it.

Bite #1: Medicaid take-up jumped in expansion states

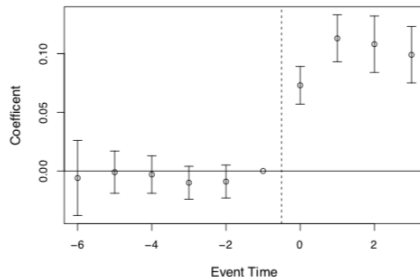


(a) Medicaid Eligibility

Did the policy actually move people onto Medicaid? Yes — enrollment rose sharply in expansion states post-2014, flat in non-expansion.

Without bite, there's no story. The mandate has to actually do something before we can attribute downstream changes to it.

Bite #2: any-insurance coverage rose

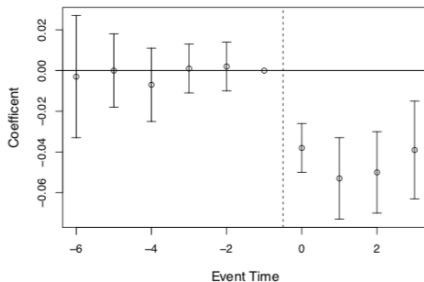


(b) Medicaid Coverage

Did the new Medicaid enrollees actually gain coverage they didn't have? Or did they just shift from private to Medicaid?

Any-insurance coverage rose — the expansion brought in the previously uninsured, not the already-insured.

Bite #3: the uninsured rate fell



(c) Uninsured

*The mirror image of Bite #2. Uninsured rate dropped sharply in expansion states post-2014. Three measures (Medicaid up, any-insurance up, uninsured down) converge on the same conclusion: **the policy bit.***

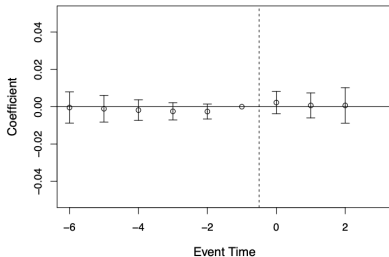
Five pieces of a strong DiD — now falsify

1. **Bite** — the policy moved what it was supposed to move. (*established.*)
2. **Event study** — pre/post coefficients around treatment.
3. **Falsification** — run a rival theory on a placebo group or outcome and try to reject *it*.
4. **Main result** — the DiD on the actual outcome.
5. **Mechanism** — the causal pathway from D to Y .

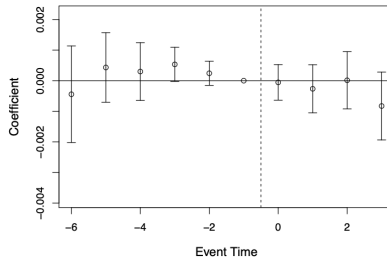
The rival: “some 2014 shock hit expansion states.” The falsification: run the same model on a group the policy shouldn’t reach.

Falsification: same model, on the 65+ population — no effect

Age 65+ in 2014



(c) Medicaid Coverage



(d) Annual Mortality

Same states, same years, same model — but run on the 65+ population. *They all have Medicare, so Medicaid expansion shouldn't touch their mortality.*

A rival theory ("some 2014 shock hit expansion states") would predict a mortality effect here. It didn't appear.

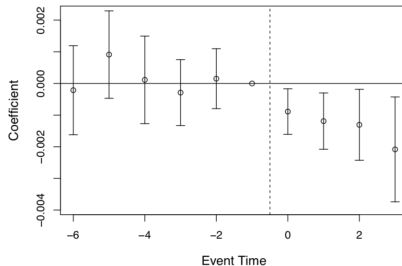
Rival rejected.

Five pieces of a strong DiD — the event study and main result

1. **Bite** — established.
2. **Event study** — pre/post coefficients around treatment.
3. **Falsification** — rival rejected.
4. **Main result** — the DiD on the actual outcome.
5. **Mechanism** — the causal pathway from D to Y .

Now we look at the outcome that matters: mortality. The event study is the main result, viewed in motion.

The main result: $\approx 9.2\%$ reduction in near-elderly mortality



- Post-2014: mortality declines in expansion states. $\approx 9.2\%$ **reduction**.
- Pre-period dots near zero, but **CIs are wide** — you cannot reject a straight line trending downward (*Rambachan & Roth 2024* sensitivity).
- The other four pieces are why the editors believed it. A critic now has to explain bite *and* falsification *and* the dynamic post-trend — not just one thing.

Five pieces of a strong DiD — and the fifth: mechanism

1. **Bite** — established.
2. **Event study** — established.
3. **Falsification** — rival rejected.
4. **Main result** — 9.2% reduction in near-elderly mortality.
5. **Mechanism** — the causal pathway from D to Y .
 - Mechanisms are the part of a result that's **transportable** — they extend across contexts where the same pathway operates.
 - Reduced-form top-line results often combine *several* mechanisms, which makes interpretation harder.
 - Here: disease-related deaths fall and compound, consistent with treated chronic conditions.

The five pieces, all five present in Medicaid

Bite	Three figures: Medicaid take-up up, any-insurance up, uninsured down
Event study	Pre-trends flat; post-trends decline in expansion states
Falsification	No effect on elderly (65+) — same states, same years, same model
Main result	$\approx 9.2\%$ reduction in mortality among the near-elderly
Mechanism	Disease-related deaths fall and compound; consistent with treated chronic conditions

*None of these alone is the case. **Together they are powerful.***

What we've done today — and tomorrow

Today:

1. DiD's empirical roots — Semmelweis, Ashenfelter.
2. DiD = calculation + assumption: the 2×2 and parallel trends.
3. Five regressions, one number.
4. Decomposing the 2×2 — the +0 trick and three failure cases.
5. Card and Krueger: parallel trends visualized.
6. Standard errors — vanilla, robust, cluster.
7. Population weighting: pick a target parameter.
8. Event studies as falsifications, by hand and via OLS.

Tomorrow: when the 2×2 stops working — covariates, staggered adoption (Goodman-Bacon, Callaway–Sant'Anna, Sun–Abraham, BJS, dCdH).